```
[4]
✓ 0s
import numpy as np
import matplotlib.pyplot as plt

from sklearn.linear_model import LinearRegression
from sklearn.preprocessing import PolynomialFeatures

# Dataset
X = np.array([[1], [2], [3], [4], [5], [6]])
y = np.array([1, 4, 9, 16, 25, 36])

# Linear Regression
linear_model = LinearRegression()

linear_model.fit(X, y)

linear_pred = linear_model.predict(X)

# Polynomial Regression
poly = PolynomialFeatures(degree=2)

X_poly = poly.fit_transform(X)

poly_model = LinearRegression()

poly_model.fit(X_poly, y)

poly_pred = poly_model.predict(X_poly)

# Prediction for X = 7
X_test = np.array([[7]])

linear_prediction = linear_model.predict(X_test)

poly_prediction = poly_model.predict(
    poly.transform(X_test)
)

print("Linear Regression Prediction:")
print(linear_prediction[0])

print("\nPolynomial Regression Prediction:")
print(poly_prediction[0])

# Smooth curve
X_curve = np.linspace(1, 6, 100).reshape(-1, 1)

linear_curve = linear_model.predict(X_curve)

poly_curve = poly_model.predict(
    poly.transform(X_curve)
)

# Graph
plt.figure(figsize=(9, 6))

plt.scatter(
    X,
    y,
    s=80,
    label="Actual Data"
)

plt.plot(
    X_curve,
    linear_curve,
    label="Linear Regression"
)

plt.plot(
    X_curve,
    poly_curve,
    label="Polynomial Regression"
)

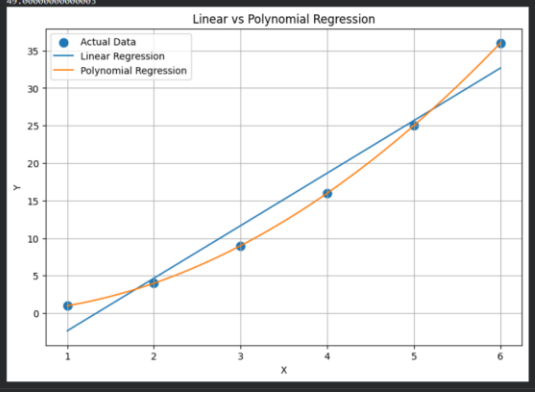
plt.xlabel("X")
plt.ylabel("Y")
plt.title("Linear vs Polynomial Regression")

plt.legend()
plt.grid(True)

plt.show()
```

```
Linear Regression Prediction:
39.66666666666667

Polynomial Regression Prediction:
49.00000000000003
```



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